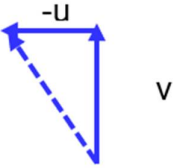


Qn	Solution
1(a)(i)	Increase in KE = Final KE - Initial KE $= \frac{mv^2}{2} - \frac{mu^2}{2}$ $= \frac{0.060}{2}(30^2 - 19^2)$ $= 16.2 \text{ J} \approx 16 \text{ J}$ (Note: incorrect to calculate $\frac{m}{2}(30 - 19)^2$)
1(a)(ii)	The change in velocity = final velocity – initial velocity  $\Delta v = \sqrt{30^2 + 19^2} = 35.5 \text{ ms}^{-1}$ Average force = $\frac{\Delta p}{\Delta t}$ $= \frac{m\Delta v}{\Delta t}$ $= \frac{0.060(35.5)}{16 \times 10^{-3}}$ $= 133 \text{ N}$ $= 130 \text{ N}$
1(b)	1. Some of the work done by the racket changes the energy of the racket itself. 2. The work done in compressing the ball by the racket is not fully recovered as KE of the ball when it rebounded, some energy is lost as heat and sound.
2(a)	Centripetal force acting on the satellite is provided by the gravitational force due